

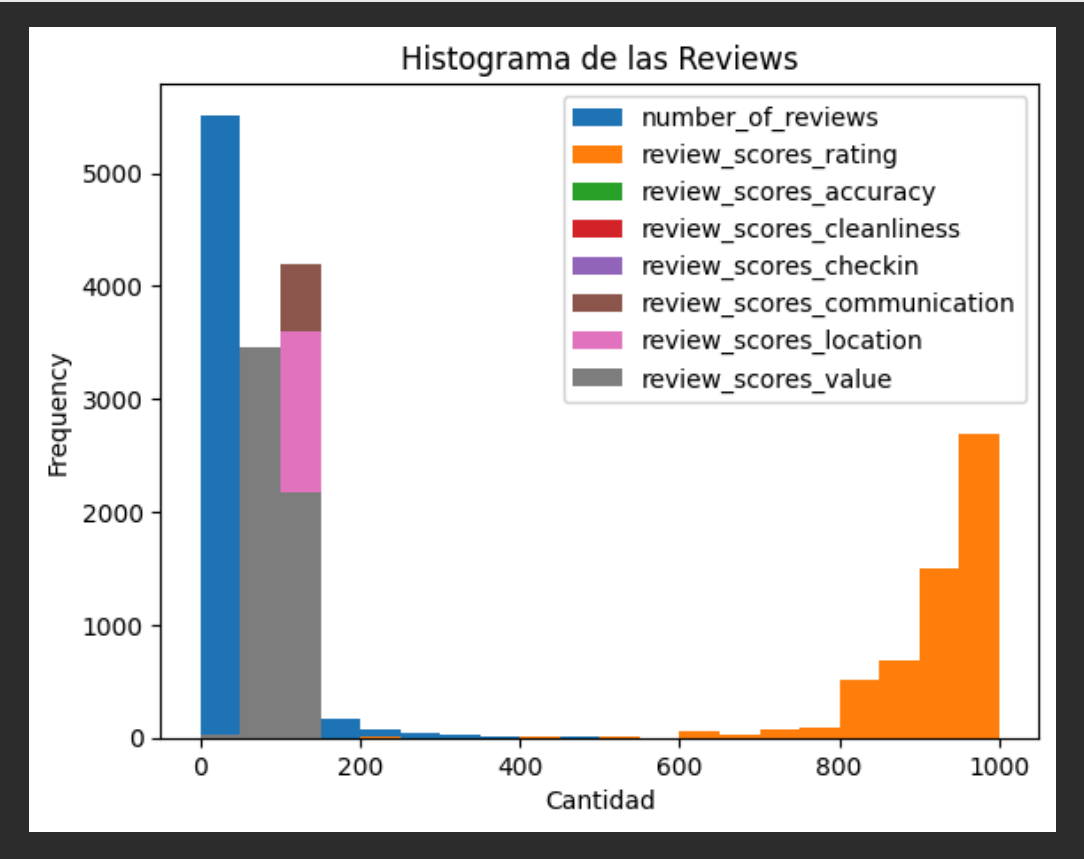
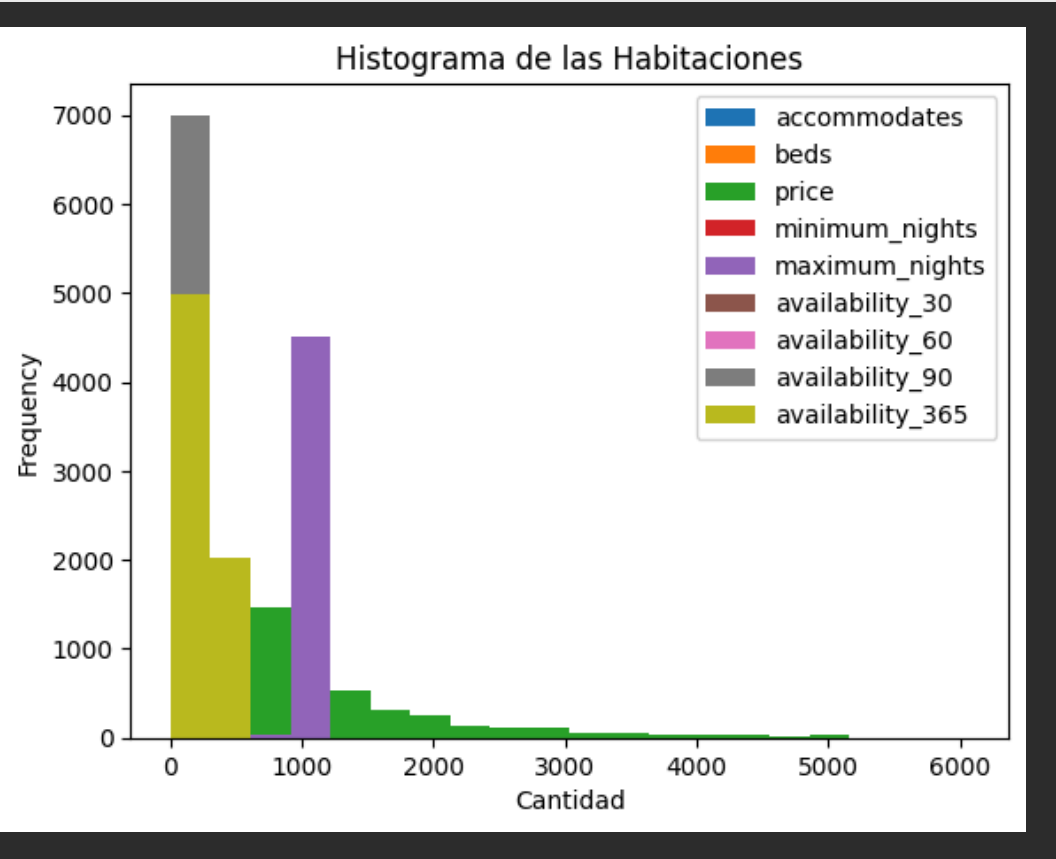
Exploración Inicial de los Datos (Exploratory Data Analysis - EDA)

Este paso es fundamental para descubrir patrones y problemas iniciales:

- Resúmenes estadísticos (media, mediana, desviación estándar, etc.).

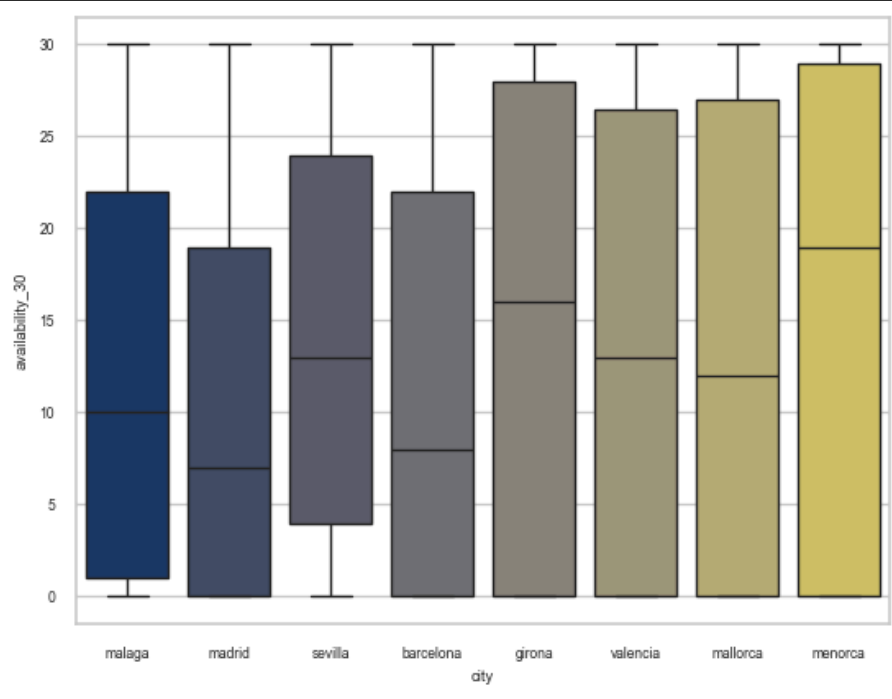
	accommodates	beds	price	minimum_nights	maximum_nights	availability_30	availability_60	availability_90	availability_365	number_of_reviews	review_scores_rating	review_scores_accuracy	review_scores_cleanliness	review_scores_checkin	review_scores_communication	review_scores_location	review_scores_value	reviews_per_month
count	7001	6998	6870	7001	7001	7001	7001	7001	7001	7001	5674	5665	5671	5660	5669	5659	5659	5747
mean	4.35	3.01	1011.1	4.42	759.47	12.28	27.42	44.31	188.13	34.13	920.23	94.54	93.2	96.29	96.43	95.31	91.46	123.04
std	2.62	2.33	845.98	10.67	497.96	11.51	22.41	33.02	129.96	59.94	85.35	9.06	9.7	7.96	7.61	7.3	9.31	153.16
min	1	0	60	1	1	0	0	0	0	0	200	20	20	20	20	20	20	1
25%	2	1	450	1	70	0	2	9	66	1	890	90	90	90	90	90	90	17
50%	4	2	750	2	1125	10	26	46	190	9	940	100	100	100	100	100	90	57
75%	6	4	1230	4	1125	24	50	76	316	40	980	100	100	100	100	100	100	177
max	29	30	6071	365	1125	30	60	90	365	588	1000	100	100	100	100	100	100	1291

- Visualizaciones iniciales para identificar tendencias o anomalías.

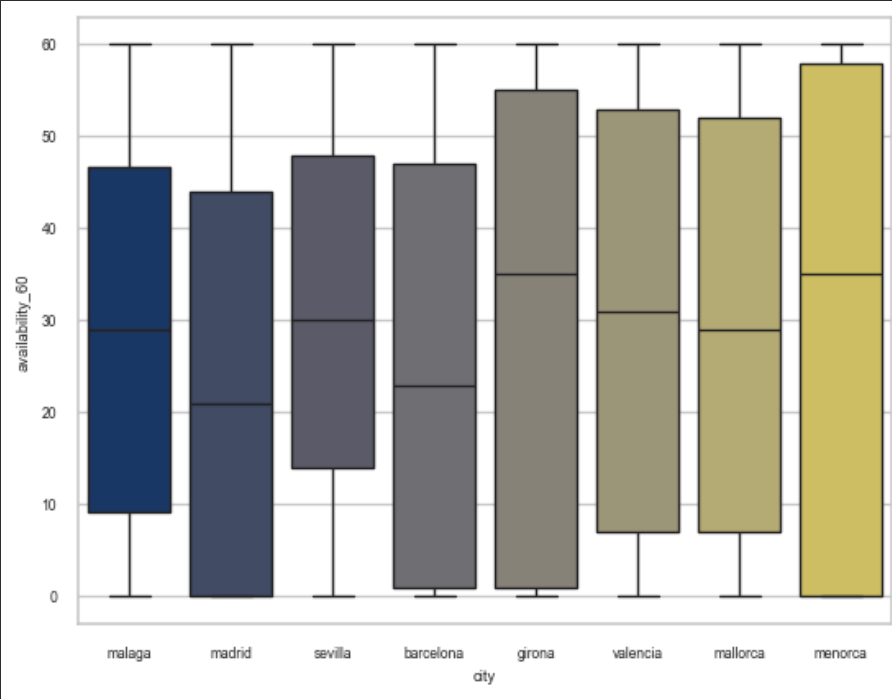


Rooms and Reviews Boxplots

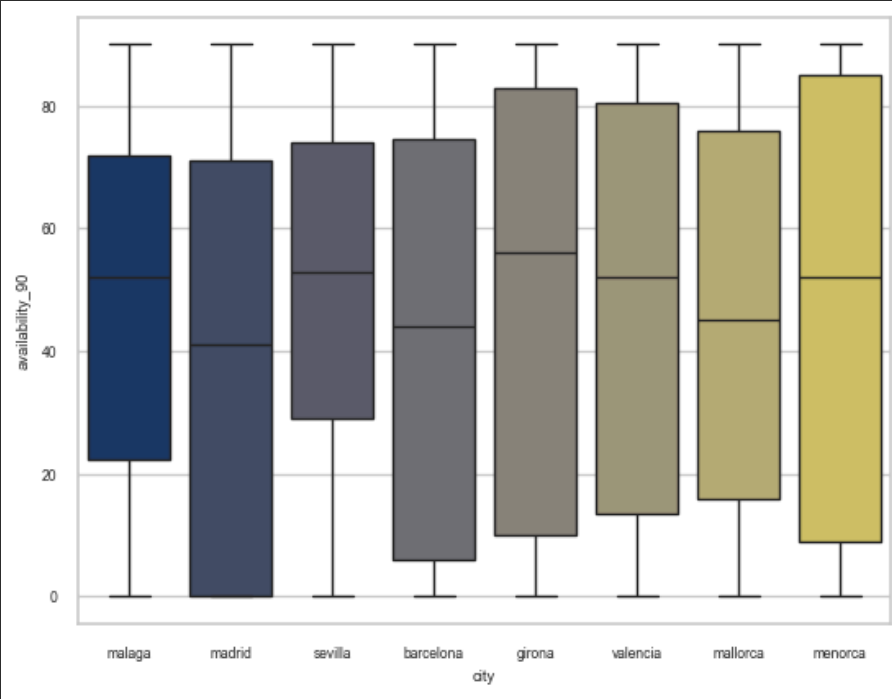
City_Rev_Availability 30



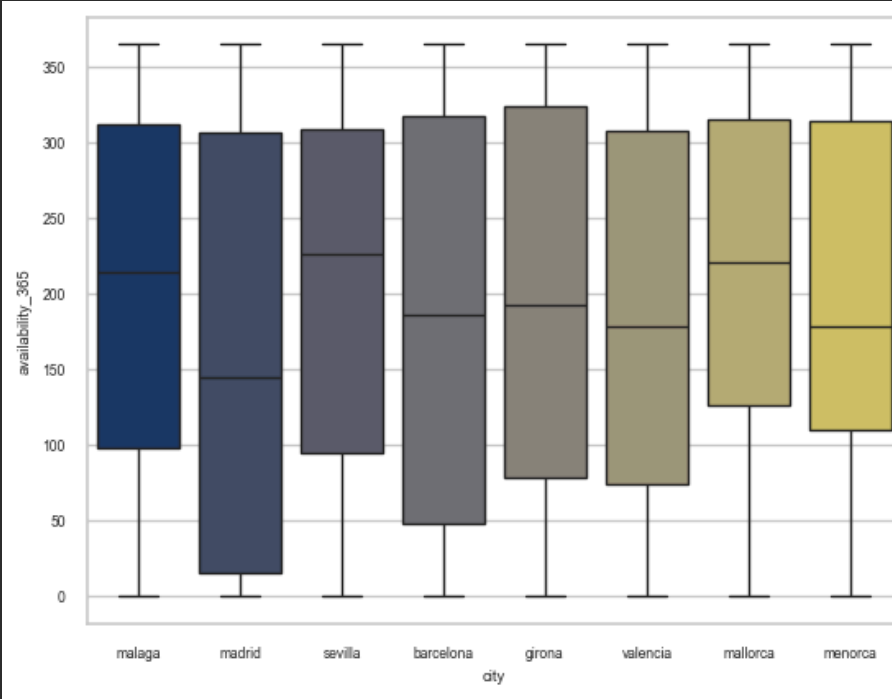
City_Rev_Availability 60



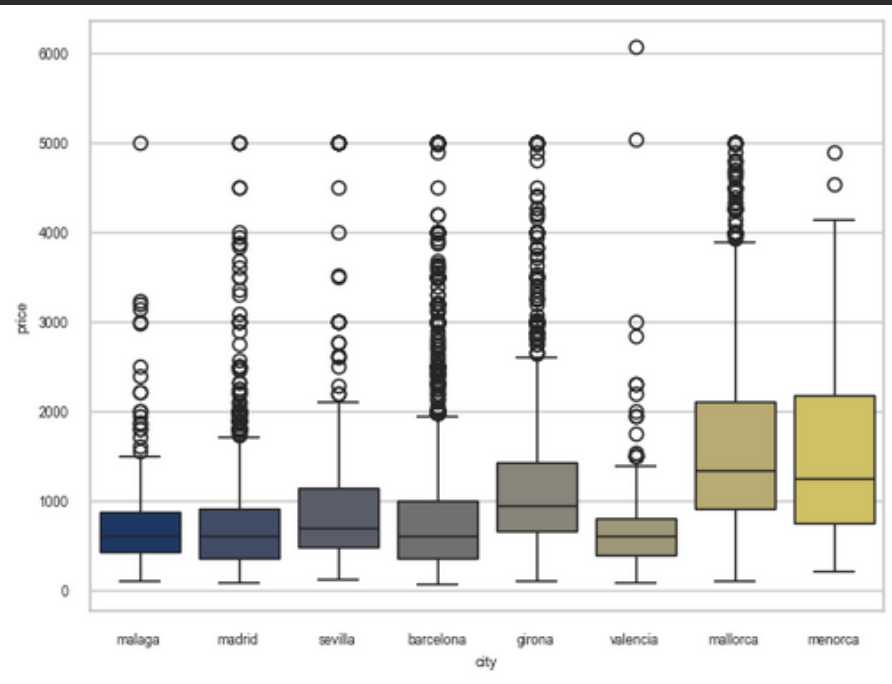
City_Rev_Availability 90



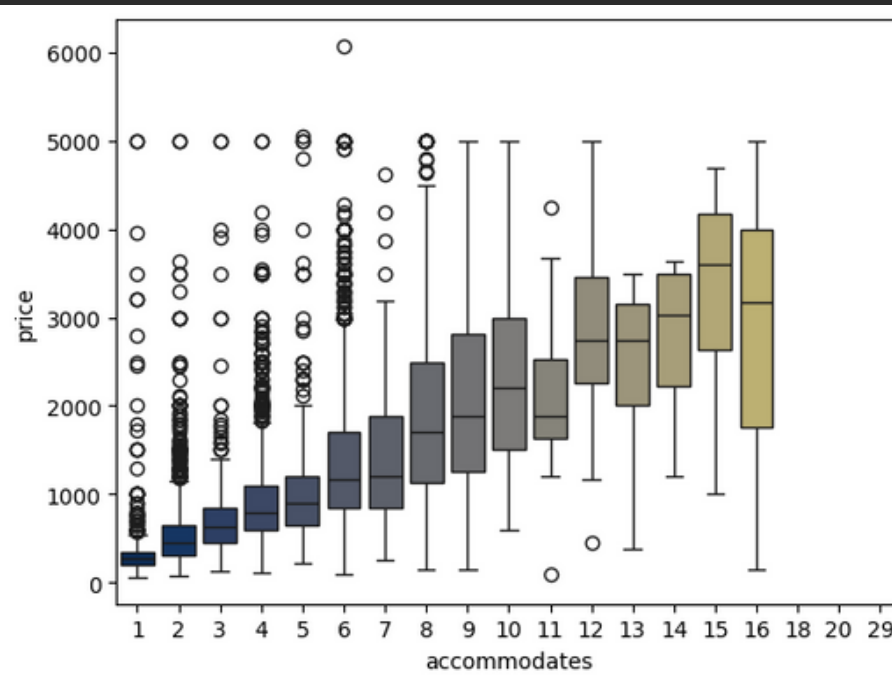
City_Rev_Availability 365



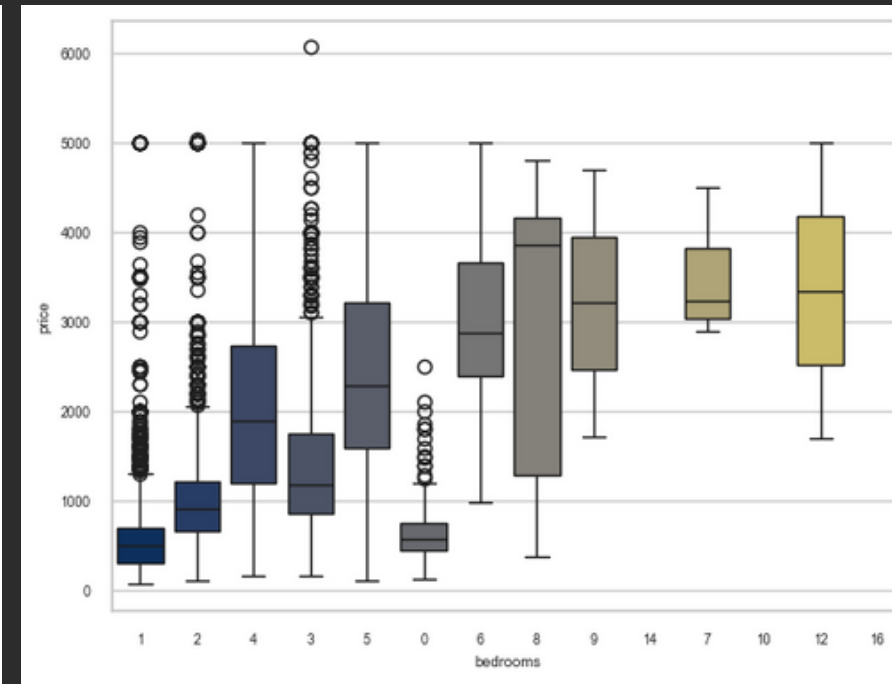
City price



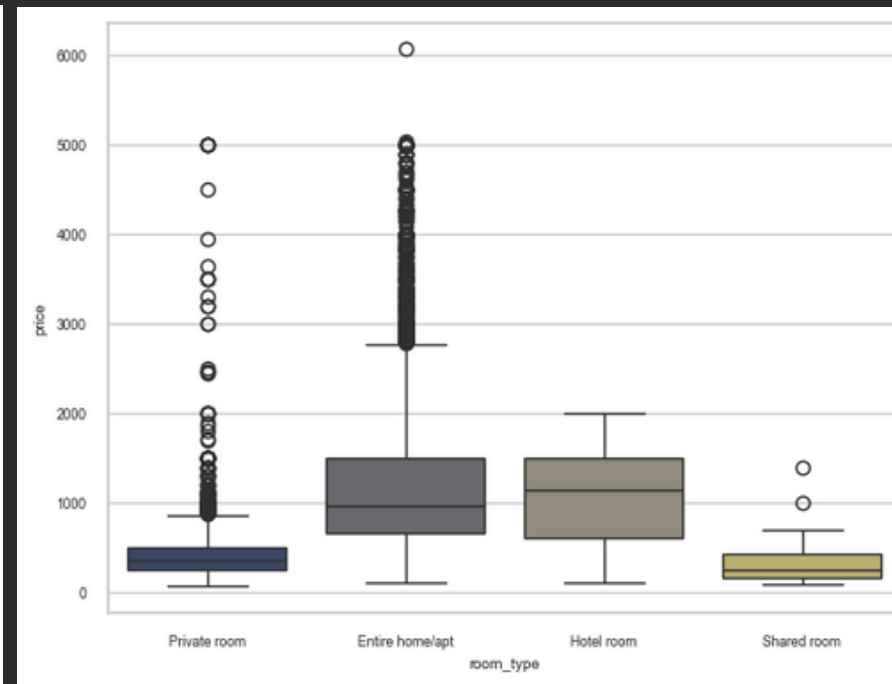
Accommodates price



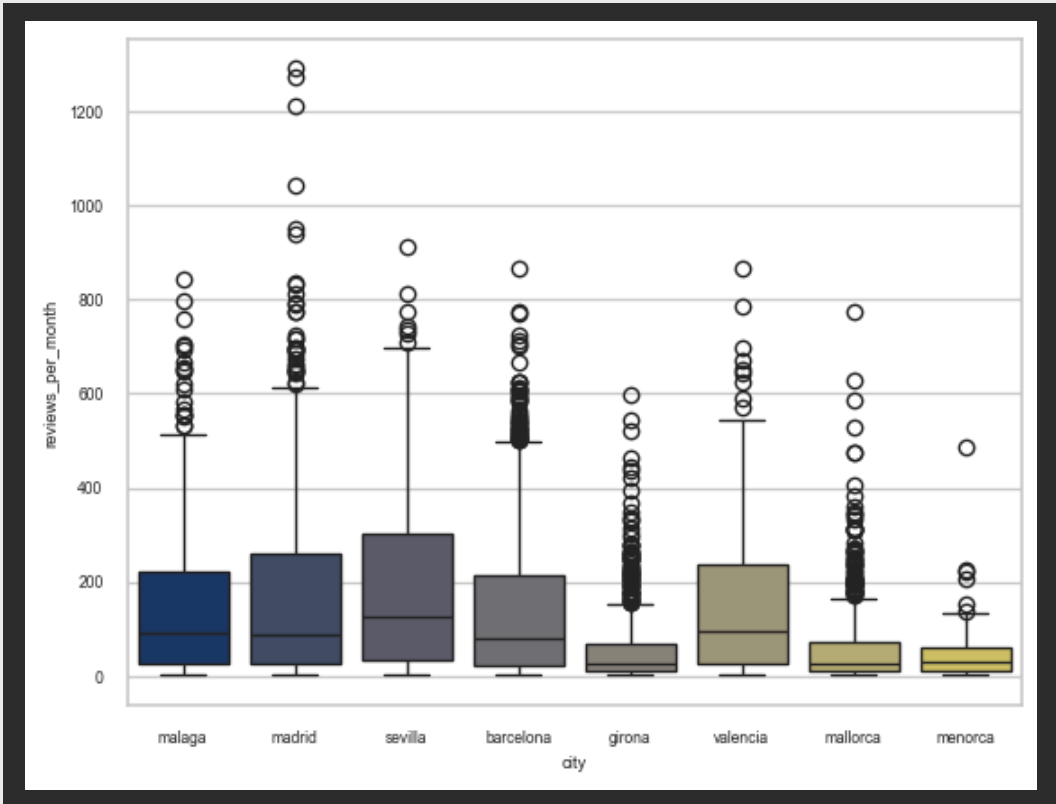
Bedrooms price



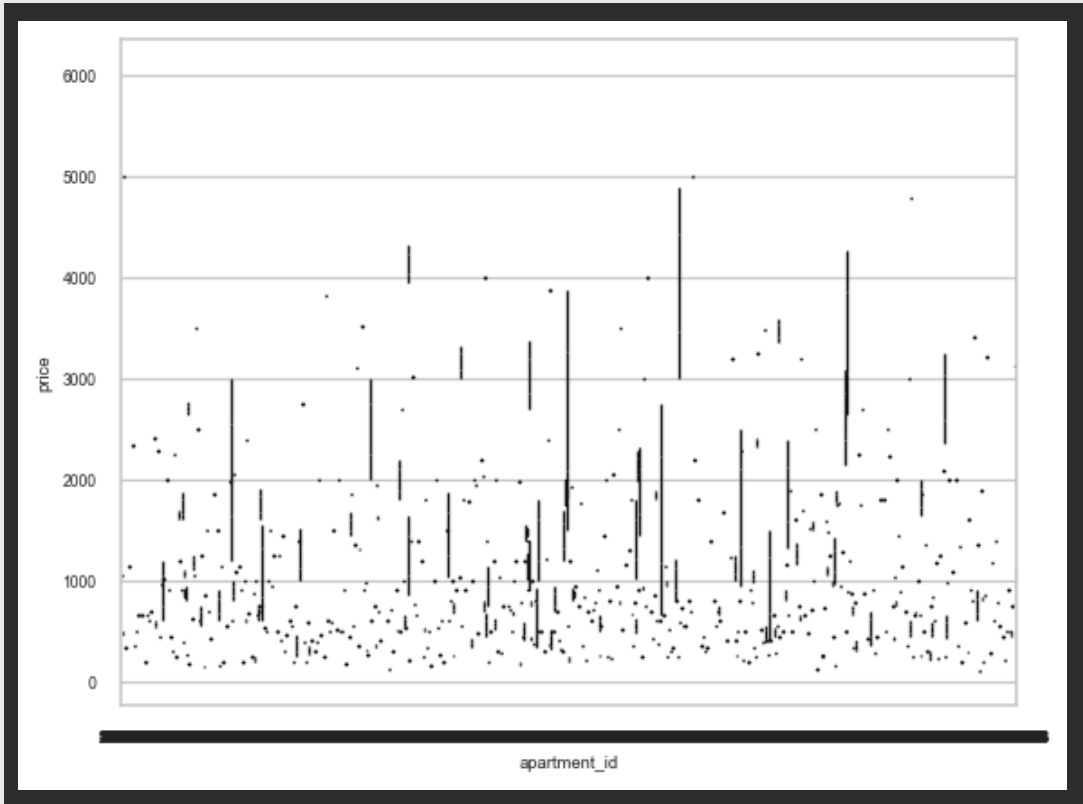
Room type price



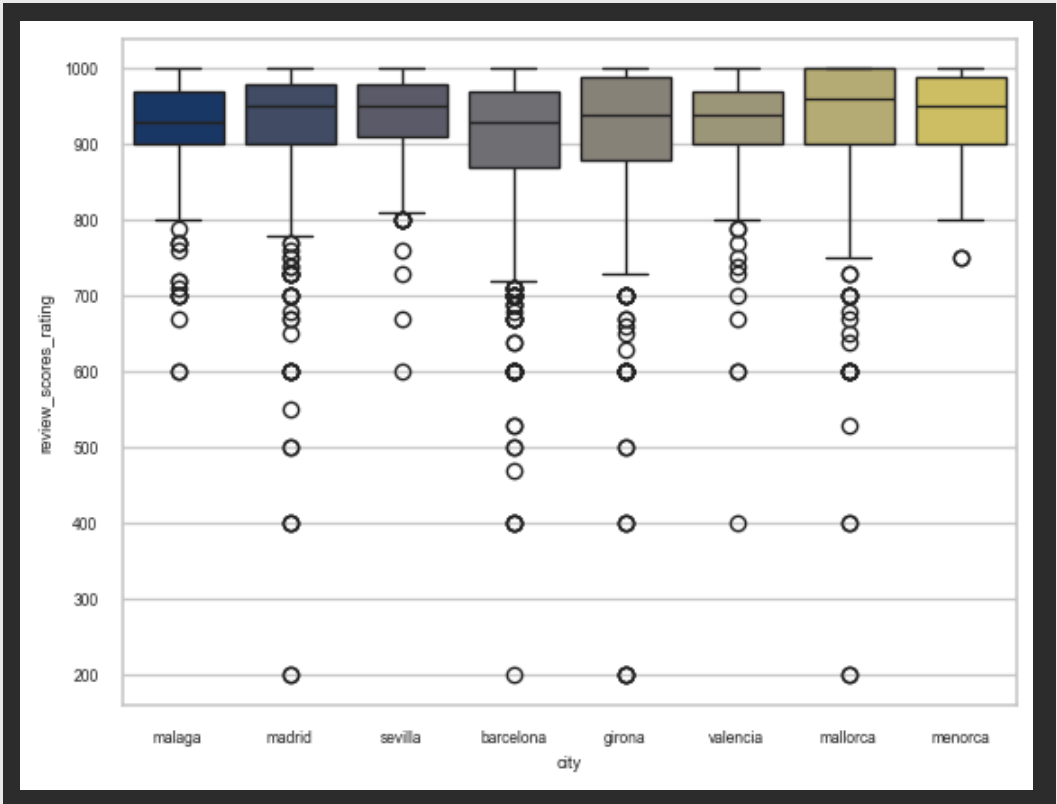
City_Rev_Reviews per month



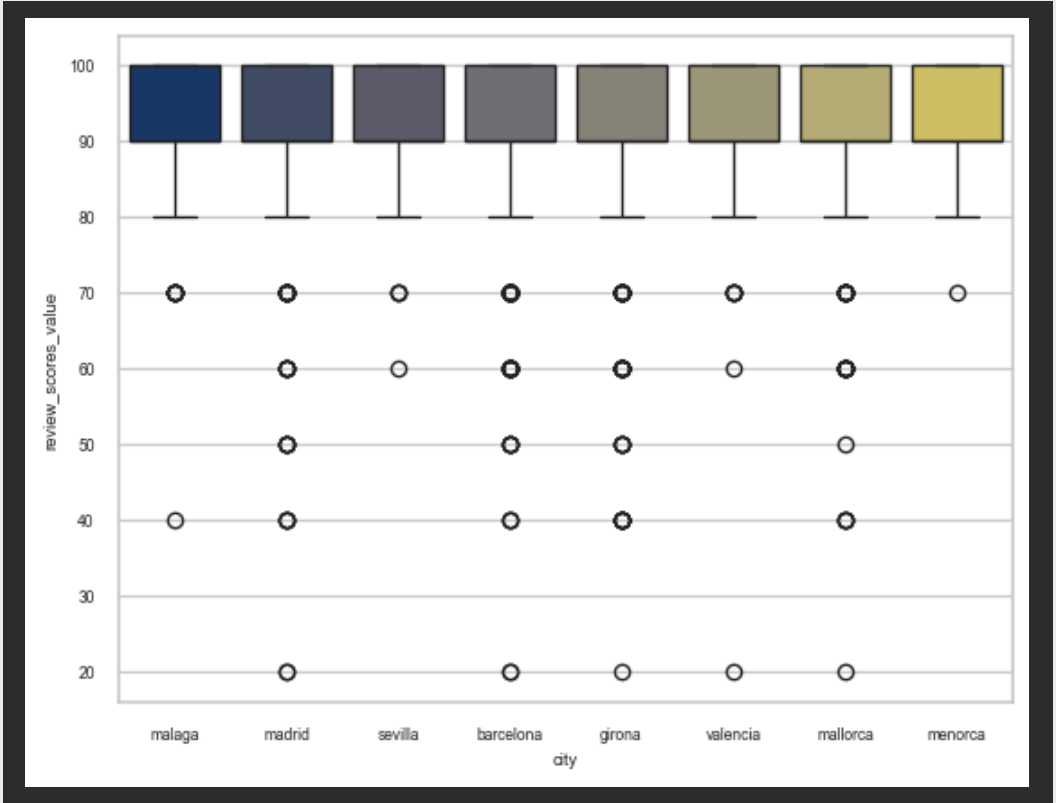
Apartment ID price



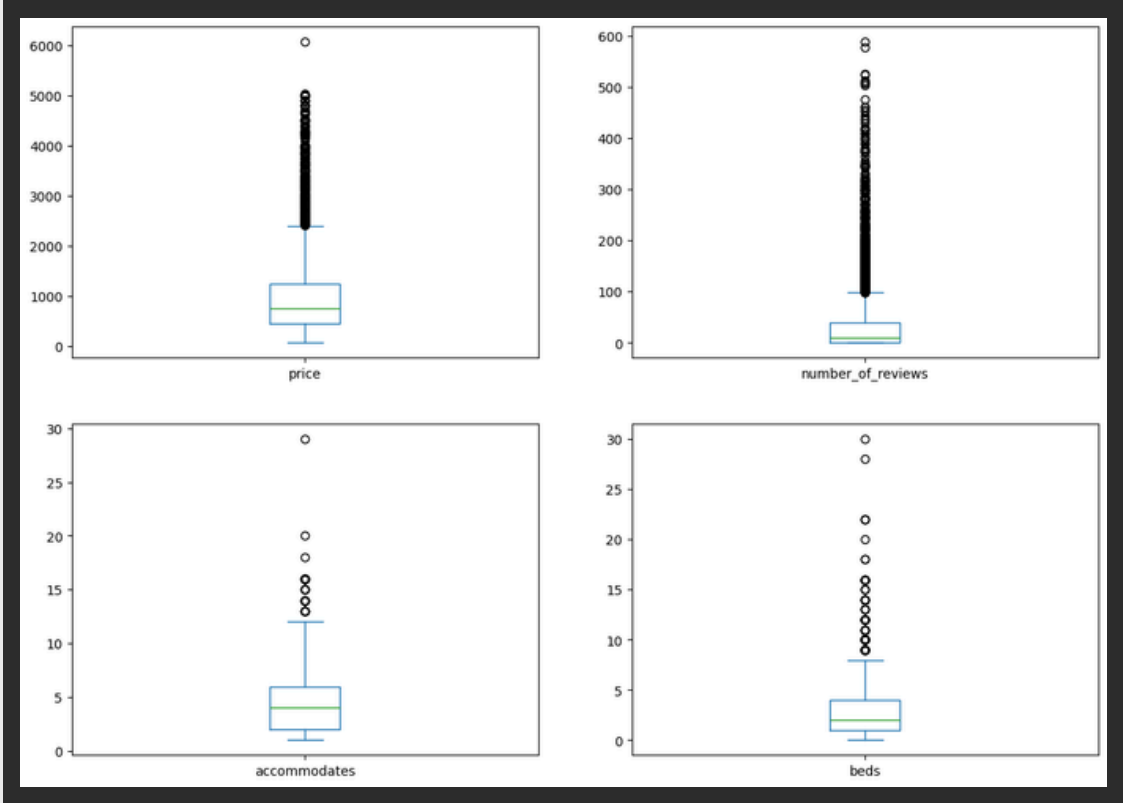
City_Rev_Scores Rating



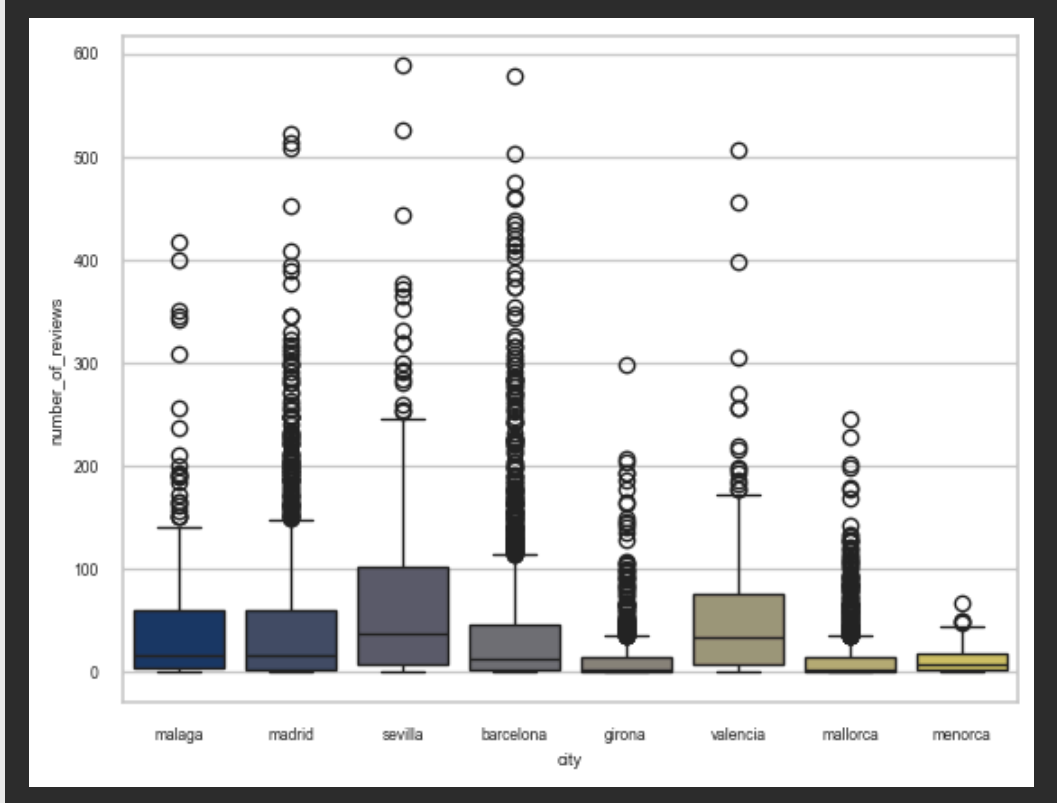
City_Rev_Scores Value



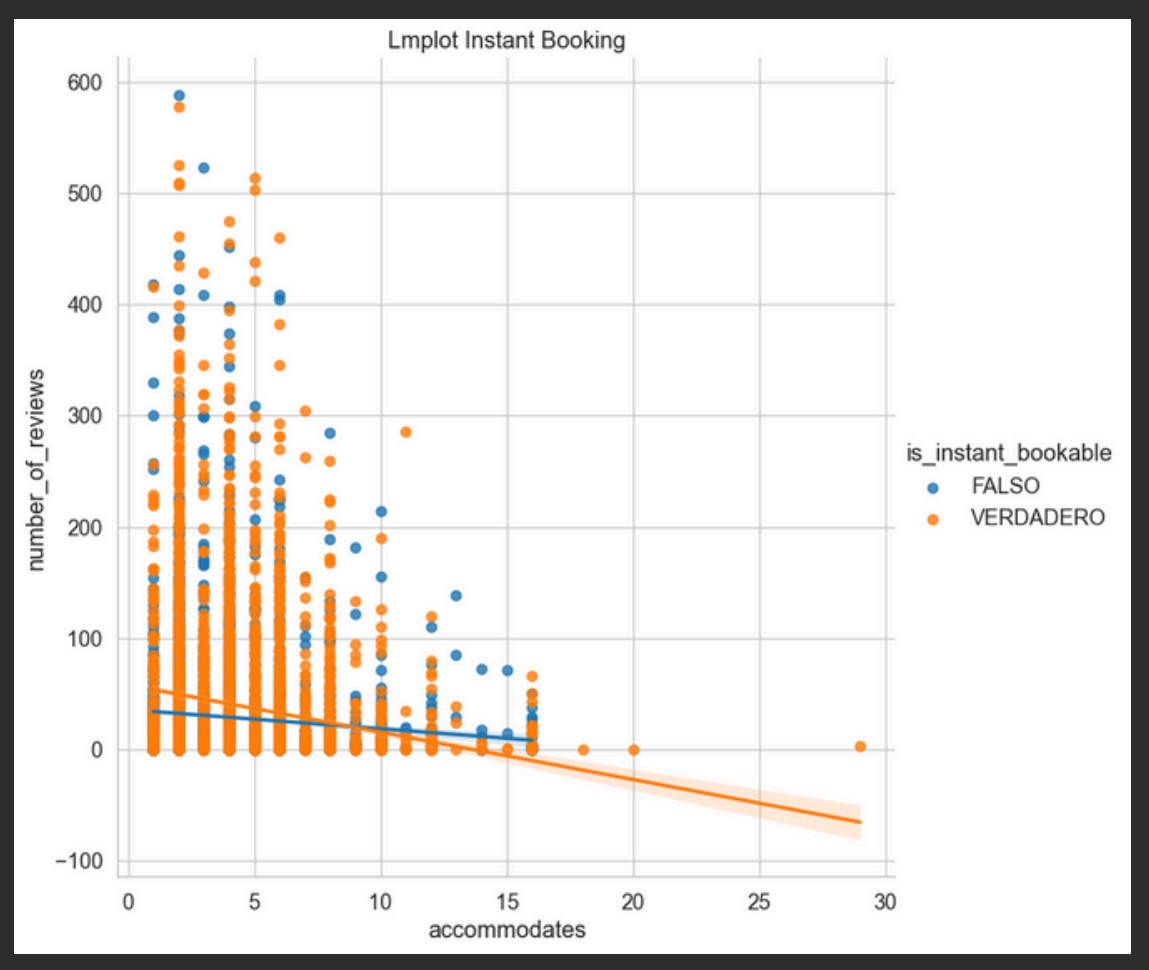
Price, Num of reviews, accommodates, beds



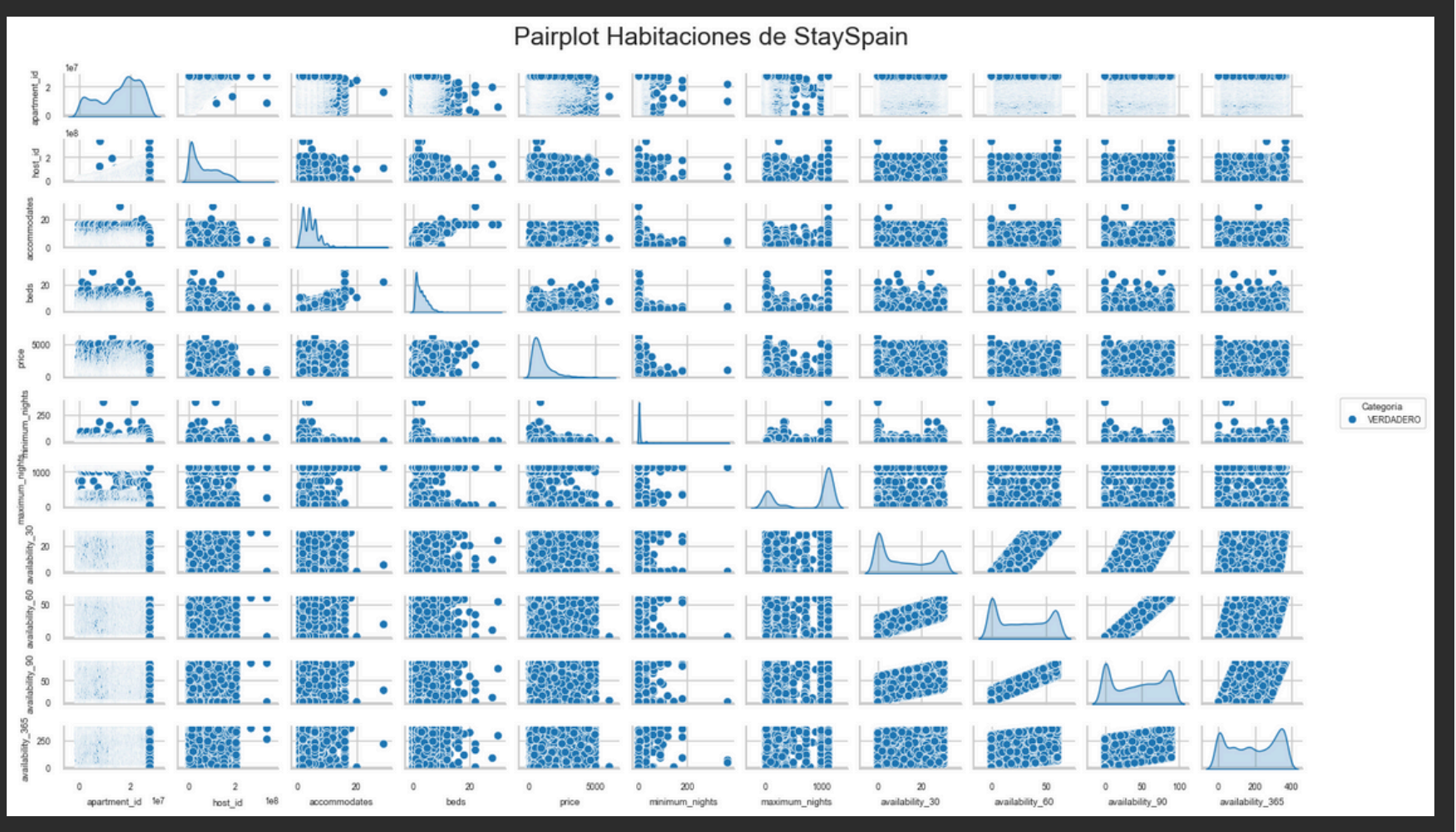
City_Rev_Number of Reviews



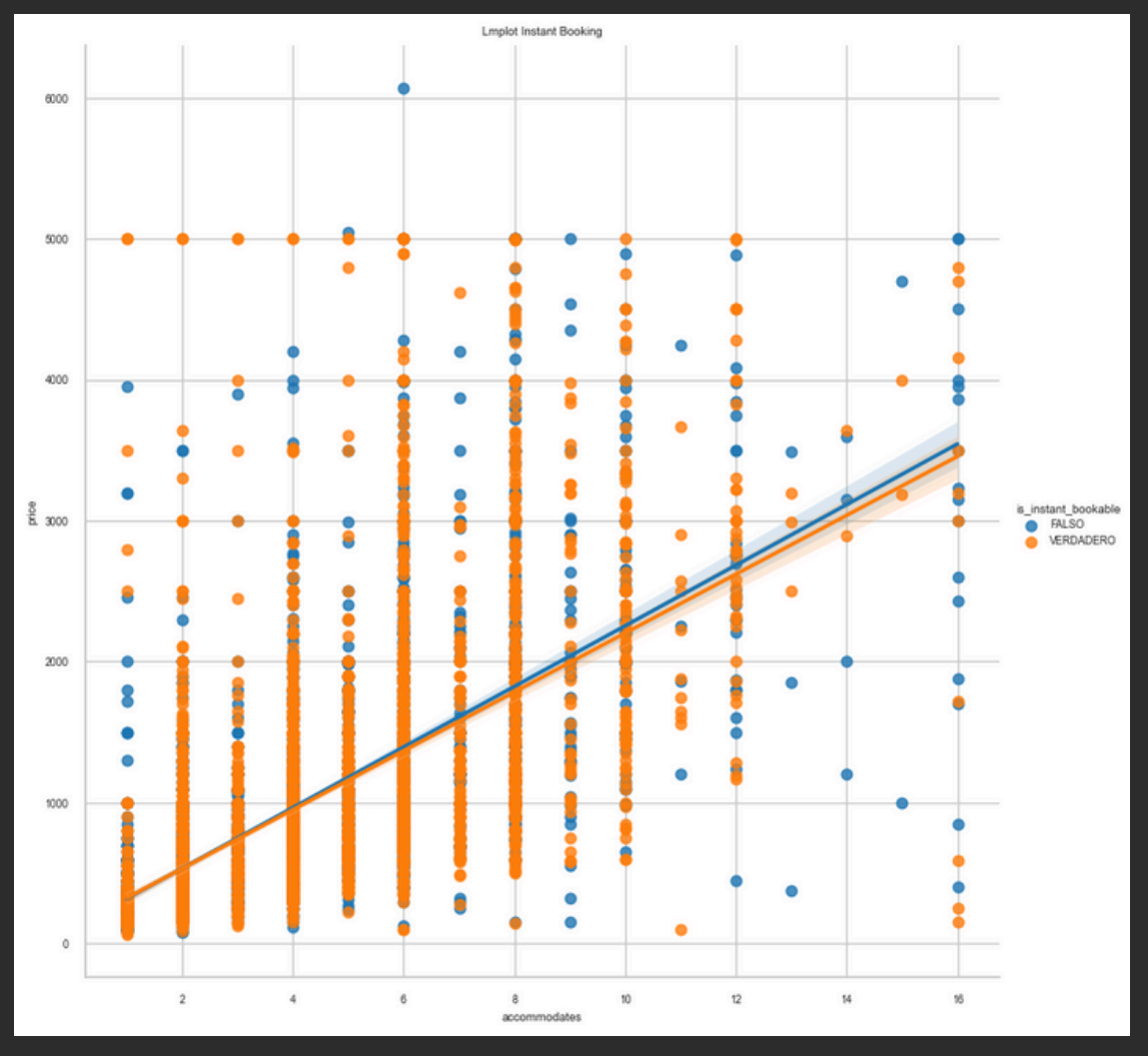
Lmplot del número de huéspedes con el **numero de reviews** y con comparativa de si tiene disponibilidad para reservar instantáneamente



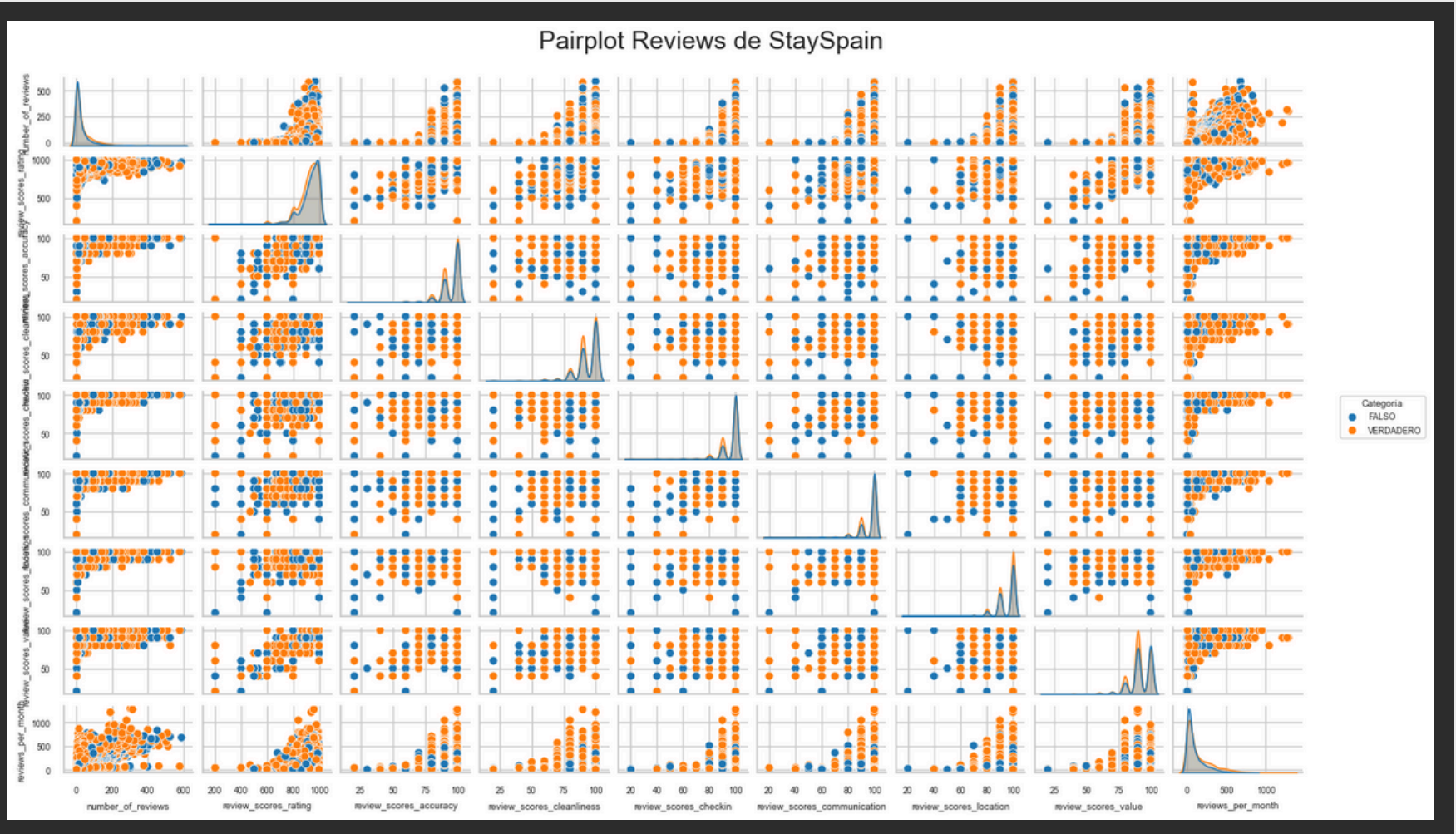
Pairplot de las variables numéricas de las habitaciones



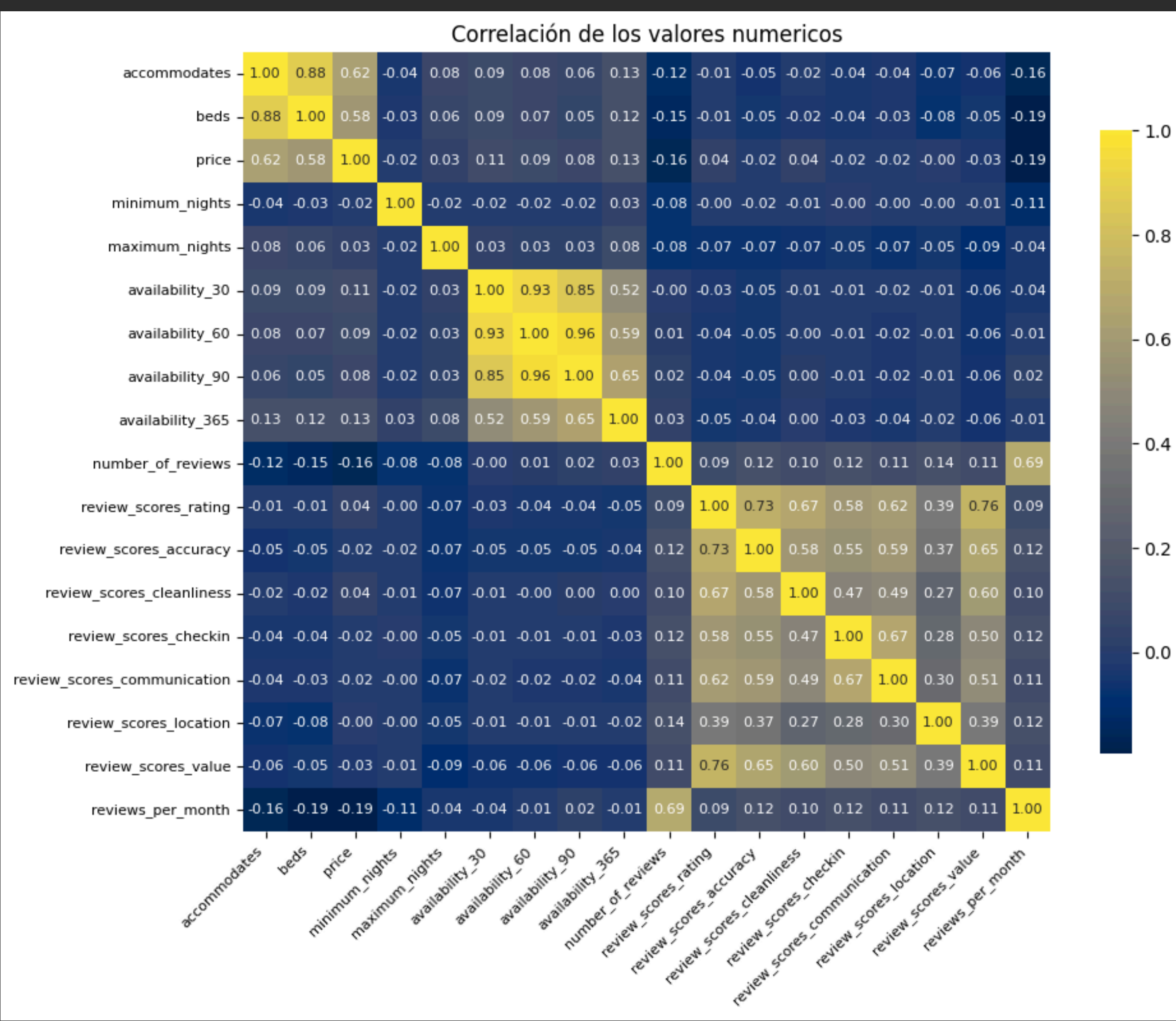
Lmplot del número de huéspedes **precio** y con comparativa de si tiene disponibilidad para reservar instantáneamente



Pairplot de las variables numéricas de las reviews



Correlación de todos los valores numéricos



Sugerencias/ Recomendaciones

- 1) Cambiar las variables IDs a varchar/objects
- 2) Las fechas cambiarlas a datetime64 para poder hacer análisis en el tiempo
- 3) Cambiar tipo de variable (varchar) a numérica (int) para las variables bathrooms, bedrooms

*Tipos por columnas:

```
|kclass 'pandas.core.frame.DataFrame'>
RangeIndex: 7001 entries, 0 to 7000
Data columns (total 35 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   apartment_id                         7001 non-null   int64
1   name                                 6998 non-null   object
2   description                           6972 non-null   object
3   host_id                              7001 non-null   int64
4   neighbourhood_name                   7001 non-null   object
5   neighbourhood_district               4241 non-null   object
6   room_type                           7001 non-null   object
7   accommodates                         7001 non-null   int64
8   bathrooms                           6969 non-null   object
9   bedrooms                            6972 non-null   object
10  beds                                6998 non-null   float64
11  amenities_list                       6984 non-null   object
12  price                                6870 non-null   float64
13  minimum_nights                       7001 non-null   int64
14  maximum_nights                       7001 non-null   int64
15  has_availability                     6451 non-null   object
16  availability_30                       7001 non-null   int64
17  availability_60                       7001 non-null   int64
18  availability_90                       7001 non-null   int64
19  availability_365                     7001 non-null   int64
20  number_of_reviews                    7001 non-null   int64
21  first_review_date                    5747 non-null   object
22  last_review_date                     5746 non-null   object
23  review_scores_rating                 5674 non-null   float64
24  review_scores_accuracy               5665 non-null   float64
25  review_scores_cleanliness            5671 non-null   float64
26  review_scores_checkin                5660 non-null   float64
27  review_scores_communication          5669 non-null   float64
28  review_scores_location               5659 non-null   float64
29  review_scores_value                  5659 non-null   float64
30  is_instant_bookable                  7001 non-null   object
31  reviews_per_month                   5747 non-null   float64
32  country                              7001 non-null   object
33  city                                 7001 non-null   object
34  insert_date                          7001 non-null   object
dtypes: float64(10), int64(10), object(15)
```

* Cantidad de Nulos por Columna

apartment_id	0	first_review_date	1254
name	3	last_review_date	1255
description	29	review_scores_rating	1327
host_id	0	review_scores_accuracy	1336
neighbourhood_name	0	review_scores_cleanliness	1330
neighbourhood_district	2760	review_scores_checkin	1341
room_type	0	review_scores_communication	1332
accommodates	0	review_scores_location	1342
bathrooms	32	review_scores_value	1342
bedrooms	29	is_instant_bookable	0
beds	3	reviews_per_month	1254
amenities_list	17	country	0
price	131	city	0
minimum_nights	0	insert_date	0
maximum_nights	0		
has_availability	550		
availability_30	0		
availability_60	0		
availability_90	0		
availability_365	0		
number_of_reviews	0		